

STATE UNIVERSITY OF ZANZIBAR (SUZA)

School of Computing, Communication and Media Studies (SCCMS)

Department of Computer Science & Information Technology

Web Technologies

PROJECT PROPOSAL

PROJECT TITLE: University Innovation Connect (UIC): A Web-Based Collaborative Platform for Student Idea Sharing and Lecturer Feedback at SUZA

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0.1	June 2026	Khadija Ali Said Rahma Suleiman Abdalla	Initial draft

Abstract

This project proposes the development of a web-based platform called **University Innovation Connect (UIC)** to enhance student innovation and collaboration at the State University of Zanzibar (SUZA). Currently, student ideas are shared informally and are not systematically recorded, which leads to loss of valuable innovations, limited feedback from lecturers, and weak collaboration among students.

The proposed system will provide a centralized platform where students can register, submit innovative ideas, comment on other users' ideas, and receive feedback from lecturers. The system will also include a voting mechanism to help identify the most valuable and promising ideas. Lecturers will be able to guide and evaluate student innovations within the platform.

The system will be developed using React for the front-end, Spring Boot for the back-end, and PostgreSQL for database management. A structured iterative development methodology will be applied to ensure proper system design, implementation, and testing.

The expected outcome is a functional web application that improves communication between students and lecturers, preserves innovative ideas, and promotes a culture of innovation within SUZA. The system will demonstrate the practical application of modern web technologies in solving real academic challenges.

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1. Introduction

1.1 Background of the Study

Higher learning institutions increasingly encourage students to develop innovative ideas that can solve real-world and academic problems. At the State University of Zanzibar (SUZA), students frequently generate creative solutions through coursework, discussions, and group activities. However, most of these ideas are shared informally and are not systematically recorded or managed.

Currently, there is no centralized web-based platform at SUZA that allows students and lecturers to share, document, and collaboratively develop innovative ideas. This leads to loss of valuable ideas, limited feedback from lecturers, and weak collaboration among students. In many cases, similar ideas are repeated across different academic projects due to the absence of a shared innovation repository.

As a result, there is a need for a structured digital system that supports idea sharing, feedback, and collaboration between students and lecturers in an organized manner. This project therefore proposes the development of a web-based platform to address this gap and improve innovation management within the university.

1.2 Problem Statement

Despite the presence of innovative thinking among students at the State University of Zanzibar (SUZA), there is no centralized system that supports the structured sharing, documentation, and development of these ideas. As a result, most student innovations are communicated informally through discussions, messaging platforms, or classroom interactions, where they are easily lost or forgotten.

This lack of a formal platform leads to several challenges, including loss of valuable ideas, limited collaboration between students and lecturers, insufficient feedback on student innovations, and repetition of similar academic projects. Additionally, there is no effective mechanism for tracking, evaluating, or improving student ideas over time.

Therefore, there is a need to develop a web-based system that provides a centralized and organized environment for students and lecturers to share ideas, collaborate, and enhance innovation within the university.

1.3 Aim of the Project

The aim of this project is to develop a web-based platform that enables students and lecturers at the State University of Zanzibar (SUZA) to share, manage, and collaboratively develop innovative ideas in a structured and centralized environment.

1.4 Specific Objectives

The specific objectives of this project are:

- To design and develop a secure user authentication system that allows students and lecturers to register and log into the platform.
- To implement a system that enables students to create, post, and manage innovative ideas within the platform.
- To develop a commenting and feedback system that allows lecturers and students to provide constructive input on shared ideas.
- To design and implement a voting mechanism to identify and highlight the most valuable student innovations.
- To evaluate the developed system to ensure it meets user requirements and supports effective collaboration.

1.5 Research / Project Questions

The project is guided by the following questions:

- How can a web-based system improve the sharing and management of student innovative ideas at SUZA?
- What features are required in a platform that supports collaboration between students and lecturers in innovation activities?
- How can feedback and voting mechanisms enhance the quality and visibility of student ideas within the system?

1.6 Scope and Limitations

1.6.1 Scope

The University Innovation Connect (UIC) system will focus on providing a web-based platform that enables students and lecturers at the State University of Zanzibar (SUZA) to:

- Register and log into the system securely.
- Post innovative ideas and problem-solving suggestions.
- View, comment on, and discuss shared ideas.
- Provide feedback and academic guidance on student innovations.
- Vote or like ideas to highlight the most valuable contributions.

1.6.2 Limitations

This project will have the following limitations:

- It will not include a mobile application version.
- It will not implement artificial intelligence or machine learning features.
- It will not support real-time chat or video communication.
- It will not integrate with external university systems such as student registration systems.
- It will be limited to a prototype-level implementation within the project timeline.

1.7 Significance of the Project

The University Innovation Connect (UIC) system is expected to significantly improve how students and lecturers at the State University of Zanzibar (SUZA) interact in relation to innovation and idea development.

For students, the system will provide a structured platform to share and develop innovative ideas, receive feedback, and collaborate with peers, which will enhance their creativity and academic engagement. For lecturers, the system will simplify the process of monitoring student innovations, providing guidance, and identifying promising ideas for further development.

For the university as a whole, the system will help preserve student-generated ideas, reduce repetition of similar academic projects, and promote a culture of innovation and collaboration. Additionally, the project will serve as a practical demonstration of modern web development technologies, contributing to the academic and technical growth of the developers.

2. Literature Review and existing Systems

2.1 Review of Related Work

Several web-based systems exist that support communication, collaboration, and idea sharing in different environments.

One example is **Google Classroom**, which is widely used in education to share learning materials and manage assignments between teachers and students. However, it is mainly focused on classroom activities and does not specifically support structured innovation or idea development.

Another example is **GitHub Discussions**, which allows developers to share ideas and discuss projects. While it is useful for collaboration in software development, it is not designed for academic environments or for lecturer-guided feedback on student innovations.

ResearchGate is also a platform used by researchers to share academic work and collaborate on research. However, it is mainly intended for researchers and not suitable for undergraduate students who need a simple system for sharing early-stage ideas.

In general, these systems show the importance of sharing and collaboration, but they do not fully support structured idea submission, feedback, and voting in a university setting. This creates a gap that the University Innovation Connect (UIC) system aims to fill.

2.2 Comparison of Existing Systems

System / Source	Key Features	Limitations / Gap
Google Classroom	Assignment sharing, announcements, feedback	Not designed for innovation sharing or idea development
GitHub Discussions	Idea sharing, collaboration, issue tracking	Mainly for developers, not suitable for academic idea management
ResearchGate	Academic sharing and research collaboration	Focused on researchers, not undergraduate student ideas
Proposed System (UIC)	Idea posting, commenting, voting, lecturer feedback	Specifically designed for SUZA students and lecturers to manage innovation ideas

2.3 Summary of the Gap

Existing systems such as Google Classroom, GitHub Discussions, and ResearchGate provide useful features for communication and collaboration. However, they are not specifically designed to support structured idea sharing, feedback, and voting within a university environment.

There is a clear gap in providing a simple and centralized platform that allows students and lecturers at SUZA to collaboratively manage innovative ideas from submission to evaluation.

The University Innovation Connect (UIC) system is designed to fill this gap by offering a dedicated platform for idea sharing, feedback, and academic collaboration within the university context.

3. The Proposed System

3.1 Overview

The University Innovation Connect (UIC) is a web-based platform designed to support innovation and collaboration between students and lecturers at the State University of Zanzibar (SUZA). The system provides a centralized environment where users can register, log in, and interact through sharing and discussing innovative ideas.

Students will be able to post ideas or solutions to academic and real-life problems, while other users can view, comment, and vote on these ideas. Lecturers will have the ability to provide feedback and guide students in improving their innovations.

The system is designed to improve communication, encourage collaboration, and create a structured environment for managing student ideas within the university.

3.2 Main Functionalities

The University Innovation Connect (UIC) system will provide the following main functionalities:

- User registration and login for students, lecturers, and administrators.
- User profile management.
- Ability for students to create and post innovative ideas.
- Viewing and browsing ideas submitted by other users.
- Commenting on ideas to support discussion and collaboration.
- Lecturer feedback and guidance on student ideas.
- Voting or liking system to highlight the most valuable ideas.
- Basic administration features for managing users and content.

3.3 High-Level Architecture

The University Innovation Connect (UIC) system will follow a **three-tier client–server architecture**, consisting of the presentation layer, application layer, and data layer.

The **presentation layer** will be developed using React.js, which provides the user interface for students, lecturers, and administrators. It allows users to interact with the system through web pages such as login, idea posting, commenting, and voting.

The **application layer** will be developed using Spring Boot, which handles the business logic of the system. It processes user requests, manages authentication, and communicates between the front-end and the database using RESTful APIs.

The **data layer** will use PostgreSQL to store all system data, including user accounts, ideas, comments, and votes in a structured and secure manner.

This architecture ensures clear separation of concerns, making the system easier to develop, maintain, and scale.

3.4 Proposed Technology Stack

Layer	Proposed Technology	Justification
Front-end	HTML5, CSS3, JavaScript, React	To create a responsive, interactive, and user-friendly web interface for students and lecturers.
Back-end	Spring Boot (Java)	To develop secure, scalable, and maintainable RESTful APIs for handling system logic and user requests.
Database	PostgreSQL	To securely store user accounts, ideas, comments, votes, and other system data with high reliability.
Hosting / Other	GitHub, Git, Postman, REST API	GitHub and Git will be used for version control and collaboration, Postman for testing APIs, and REST API for communication between the front-end and back-end.

4. Methodology

4.1 Software Development Methodology

This project will use the **Agile Software Development methodology**. Agile is suitable for this project because it allows the system to be developed in small, manageable increments while continuously improving through feedback and testing.

The development will be carried out in short iterations (sprints), where each sprint focuses on implementing specific features such as user authentication, idea posting, commenting, and voting. After each iteration, the system will be reviewed and improved based on testing results and feedback.

Agile is appropriate for this project because it supports flexibility, continuous improvement, and faster delivery of functional components, making it ideal for a student-based web development project.

4.2 Requirements Gathering Techniques

The following techniques will be used to gather system requirements for the University Innovation Connect (UIC) project:

- **Interviews:** Informal interviews will be conducted with students and lecturers at SUZA to understand their needs regarding idea sharing, feedback, and collaboration.
- **Observation:** The current methods used by students to share ideas (such as WhatsApp groups and classroom discussions) will be observed to identify existing challenges.
- **Document Analysis:** Existing academic guidelines and project reports will be reviewed to understand how student innovations are currently managed.

These techniques will help ensure that the system is designed based on real user needs and actual problems within the university environment.

4.3 System Analysis and Design Tools

The following tools and techniques will be used to analyze and design the University Innovation Connect (UIC) system:

- **Use Case Diagrams (UML):** To represent the interaction between users (students, lecturers, administrators) and the system functionalities.
- **Entity Relationship Diagram (ERD):** To model the database structure, including entities such as users, ideas, comments, and votes.
- **Class Diagram (UML):** To represent the structure of the system, including classes, attributes, and relationships between objects in the system.

These tools will help in understanding system requirements clearly and ensuring proper system design before implementation.

4.4 Testing Approach

The University Innovation Connect (UIC) system will be tested to ensure that all functionalities work correctly and meet user requirements.

The following testing methods will be used:

- **Unit Testing:** Individual components such as user authentication, idea posting, and voting will be tested separately to ensure they work correctly.
- **Integration Testing:** Different system modules (frontend, backend, and database) will be tested together to ensure proper communication and data flow.
- **System Testing:** The complete system will be tested as a whole to verify that all features function as expected.
- **User Acceptance Testing (UAT):** Final testing will be conducted with potential users (students and lecturers) to ensure the system meets their needs and expectations.

4.5 Deliverables

The following are the expected deliverables of this project:

- Software Requirements Specification (SRS)
- Software Design Description (SDD)
- A fully functional web-based application (University Innovation Connect - UIC)
- Source code hosted on GitHub
- Test report documenting system testing and results
- User manual for system usage
- Final project report and presentation slides

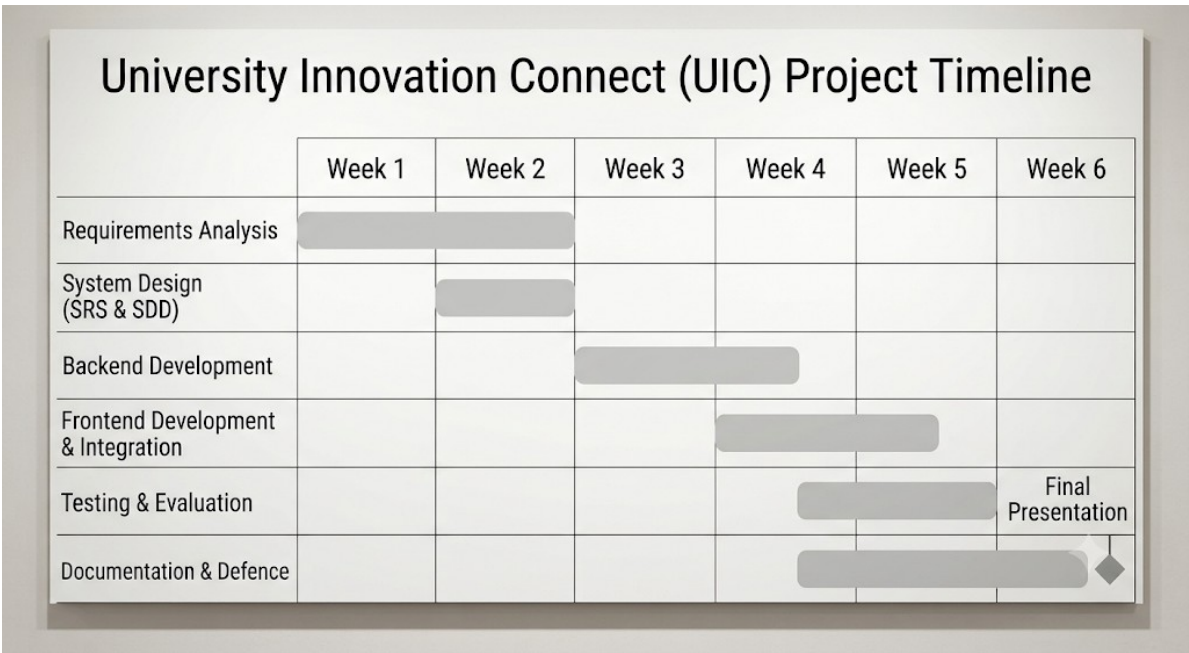
5. Project Plan and Schedule

5.1 Work Breakdown and Timeline

Phase	Activity	Approx. Duration
1	Requirements gathering & analysis	1 week
2	System design (SRS, SDD)	1 week
3	Implementation / coding	1 week
4	Testing & evaluation	1 week
5	Documentation & defence	1 week

5.2 Gantt Chart

The following Gantt chart illustrates the timeline and progression of the University Innovation Connect (UIC) project across a 5-week development period.



5.3 Resource Requirements

The following resources will be required for the development and implementation of the University Innovation Connect (UIC) system:

Resource	Purpose	Source / Cost
Laptop	System development, coding, and testing	Personal / Student-owned
Internet Access	Research, collaboration, and software downloads	20,000 TZS (estimated)
Visual Studio Code	Code editor for frontend and backend development	Free
React.js	Front-end development framework	Free and open-source
Spring Boot	Back-end development framework	Free and open-source
PostgreSQL	Database management system	Free and open-source
Git & GitHub	Version control and project hosting	Free
Postman	API testing and debugging	Free
Lecturer Supervision	Guidance and project evaluation	University support

5.4 Risks and Mitigation

Risk	Likelihood	Impact	Mitigation
Limited time for project completion	Medium	High	Follow Agile methodology and divide work into small weekly tasks
System bugs and errors during development	High	Medium	Perform continuous testing and debugging during each iteration
Lack of experience with some technologies	Medium	Medium	Practice through small prototypes and guided learning
Technical difficulties in backend or API development	Medium	High	Use online documentation, tutorials, and lecturer support

6. References

- [1] React Documentation, “React – A JavaScript library for building user interfaces,” <https://react.dev/>, accessed June 2026.
- [2] Spring Boot Documentation, “Spring Boot Reference Guide,” <https://spring.io/projects/spring-boot>, accessed June 2026.
- [3] PostgreSQL Documentation, “PostgreSQL: The World’s Most Advanced Open Source Relational Database,” <https://www.postgresql.org/>, accessed June 2026.

AI Tools Used in This Submission

AI Tool / Service	Specifically used for
ChatGPT	Providing guidance on formatting, and academic writing style (including IEEE references)

Student Attestation

We confirm that the intellectual content of this document the analysis, design decisions, and judgments is our own work. I have declared all AI assistance above. I understand that I may be asked to explain and defend any part of this submission orally, and that being unable to do so may be treated as evidence of misconduct

Project Repository (Version Control)

Field	Value
Repository URL	https://github.com/KhadijaSaid07/University-Innovation-Connect
Date of first commit	June 2026